



CHENNAI INSTITUTE OF TECHNOLOGY, CHENNAI
Affiliated to Anna University ·
(Autonomous)
Dept. of Computer Science and
Engineering

AgriSense

ML-Based Crop Recommendation & Yield Intelligence

Analyse · Recommend · Estimate · Decide

INNOVATIVE THINKING - PBL
Think · Build · Evaluate

*Smarter Sowing,
Better Harvests*



Problem

- Most Indian holdings are small or marginal — crop choice and expected yield rest on **inherited experience**.
- Soil tests return **nutrient values, not decisions**. The gap is interpretation, not measurement.
- Shifting rainfall and repeated cropping erode the reliability of that experience.
- Prior ML work rarely reaches the farmer — it ends at a notebook, not a usable tool.



Objectives

- ✓ Preprocess two public datasets; derive yield and remove invalid and extreme records.
- ✓ Build and iteratively refine classification and regression models.
- ✓ Benchmark three models per task against a baseline, then tune the winner.
- ✓ Evaluate with accuracy, precision, recall, F1 and RMSE, MAE, R².
- ✓ Deliver the trained models through a REST API and a mobile client.



Our Approach

Two supervised tasks behind one app:

- ✓ **Classification** — recommend 1 of 22 crops from N, P, K, temperature, humidity, pH, rainfall.
- ✓ **Regression** — estimate yield per hectare from crop, season, area and conditions.
- ✓ Three models compared per task; the winner tuned by grid search and cross-validated.
- ✓ Served through a REST API and consumed by a React Native app.

Not just a model — a decision a farmer can act on.

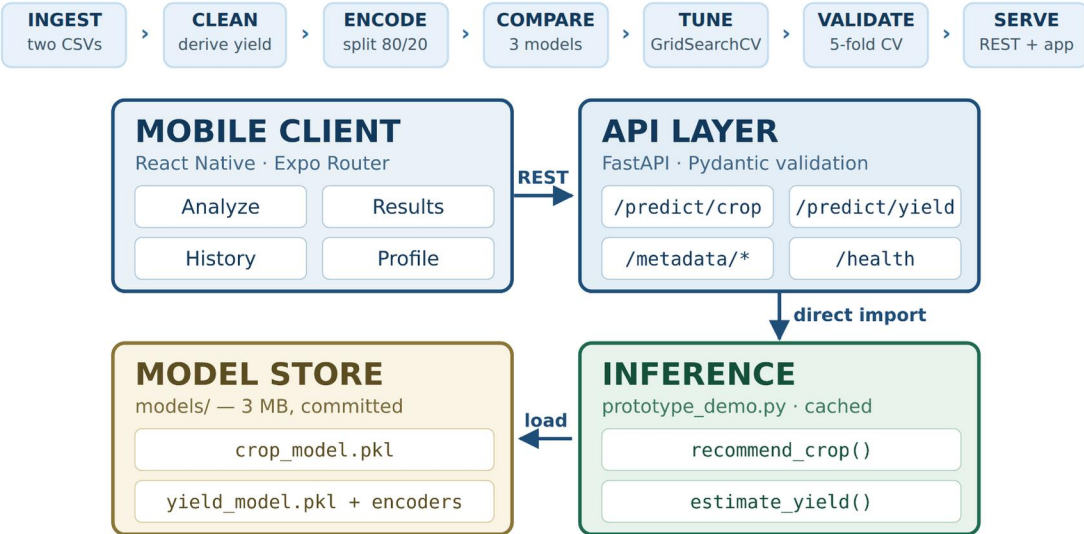


Key Features

- Crop Recommendation**
22 balanced classes
- Under a Second**
models cached in memory
- Mobile App**
React Native · Expo
- Yield Estimation**
units per hectare
- Feature Importance**
why the model chose it
- 22 Automated Tests**
9 ML + 13 API



System Architecture



Three layers, ML layer as the single source of truth. The API **imports** the inference module rather than reimplementing it — a parity test asserts endpoint output equals direct model output to 1e-5.



ML Models & Dataset

Models compared

- **Logistic / Linear Regression** — baselines
- **Random Forest** — winner, crop recommendation
- **XGBoost** — winner, yield estimation

Datasets

Crop Recommendation — 2,200 rows · 7 features · 22 balanced classes, 100 each · stratified 80/20 → 1,760 / 440

Indian Crop Production — 49,999 rows → **48,681 cleaned** · 75 crops · 6 seasons · Yield = Production ÷ Area · clipped 1st-99th pct · 15,000 sampled → 12,000 / 3,000



Prototype / Example

CROP RECOMMENDATION

N-P-K 90-42-43 · 20.9 °C · 82% humidity · pH 6.5 · 202.9 mm

RICE

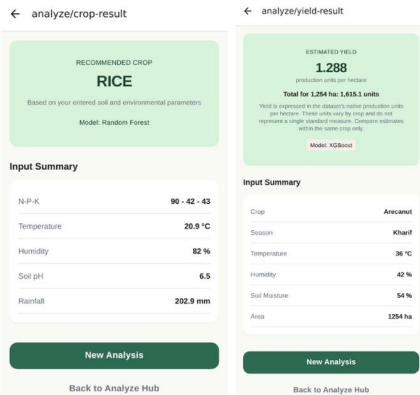
Random Forest · 22 crops analysed

YIELD ESTIMATION

Arecanut · Kharif · 1,254 ha · 36 °C · 42% · soil 54%

1.288 units / ha

XGBoost · total 1,615.1 units



Result screens from the running app against the live REST backend — values match direct model output exactly



Results & Evaluation

99.55%

Test accuracy
CV 0.9927 ± 0.0039

0.7403

R² — yield
CV 0.7428 ± 0.0254

Classification — test accuracy (440 records)

Logistic Regression		0.9727
XGBoost		0.9886
Random Forest ★		0.9955

Regression — R² (3,000 records)

Linear Regression		0.0726
Random Forest		0.6719
XGBoost tuned ★		0.7403

Confusion matrix: **438 of 440** held-out records correct. Key finding — yield is driven by crop (0.4949) + season (0.3404) = **84%** of importance, which a linear model over label-encoded integers cannot represent.

Model	Accuracy	R ²
Logistic / Linear	0.9727	0.0726
Random Forest	0.9955	0.6719
XGBoost	0.9886	0.6946
Tuned winner	0.9955	0.7403



Future Enhancements

- ✓ Train on all 48,681 cleaned rows instead of a 15,000-row sample.
- ✓ Replace the coarse integer weather columns with a meteorological API feed.
- ✓ Return class probabilities and prediction intervals as a confidence measure.
- ✓ Deploy with authentication and a restricted cross-origin policy.
- ✓ Add offline support and regional-language interface options.



Conclusion

AgriSense recommends a crop and estimates its yield from measurements a small-scale farmer can realistically obtain. A tuned **Random Forest** classifies 22 crops at **99.55%** accuracy; a tuned **XGBoost** regressor estimates yield at **R² 0.7403**. Both are served through a REST API and a mobile app, with an automated parity test asserting that what the app shows matches the model exactly.

PROJECT TEAM

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FACULTY GUIDE

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